

2.1.14 Extended Partition

An Extended Partition refers to the entry in the Master Partition Table that points to the location of the Extended Partition table. The Extended Partition is not assigned any letter.

2.1.15 Form Factor

Hard disks come in different sizes to match the systems they are intended to be used in. The 3.5-inch hard disk is what is used in all Desktop PCs. The 2.5-inch hard disk is used in laptops. Some consumer electronics products like the iPod use a 1.8-inch internal hard disk.

2.1.16 Formatting

Formatting is of two types—low-level and high-level. Low-level formatting creates the tracks and sectors that are used to store data. High-level formatting involves preparing the filesystem and creating a record of avoidable bad sectors in the partitions. All present hard disks are low-level-formatted by the manufacturer, and the user only performs a high-level format. A high-level format can be performed in two ways: a Quick Format, which erases the existing file index and creates the desired file system, and a Thorough Format, which in addition also performs a bad sector scan.

2.1.17 GMR

Giant Magneto Resistance is the phenomenon behind the technology used by the read heads in modern drives to read the data stored on disk platters. The GMR effect is a property of thin films of certain materials that experience large changes in resistance as a result of very small changes in magnetic field. This effect allows the information about changes in magnetic fields to be deciphered by detecting changes in resistance. And since even minute changes in magnetic field can be detected, these magnetised spots can be more closely packed, leading to greater storage area on a platter of the same size. The 2007 Nobel Prize for Physics was awarded to the discoverers of this effect—Albert Fert and Peter Grünberg.